

Folded Space-Time Geometry: Particles as Space-Time Cylinders and Geometric Origin of Charges

Supplementary Clarifications to Einstein-VED Theory

This document provides additional explanations and expansions to the original Einstein-VED theory, emphasizing geometric interpretations, proper time, and effects related to mass.

Serves as complement to the original work documented in Zenodo:

[The Theory That Will Reconcile Quantum Mechanics with Einstein's Relativity \(PDF in Spanish\)](#)

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1. Abstract

A conceptual framework is proposed where particles are not points, but **space-time cylinders** extended along their complete temporal history.

- All instants are real, forming a four-dimensional object in $\mathbb{R}^4$.
- Physical properties emerge from a countable series of **discrete foldings**, T_n , with $n \in \mathbb{N}$.

Interpretation of foldings:

- $n = 2$: matter/antimatter duality and electric charge
- $n = 3$: color charge and confinement

This approach unifies the **continuous** (worldline) and the **discrete** (foldings), offering an arithmetic-geometric framework for fundamental symmetries.

2. Worldline as Cylinder

Let a particle whose temporal history be described as a **space-time cylinder**, called Γ .

$$\Gamma = \{(x, y, z, t) \mid (x, y, z) \in \Sigma(t), t \in \mathbb{R}\}$$

Verbal description for transcription:

- Γ is the set of all quadruples (x, y, z, t) , where t spans all real numbers.
- For each instant t , the particle occupies the spatial section $\Sigma(t)$ in three-dimensional space.
- Conceptually, $\Sigma(t)$ represents the particle's location in $\mathbb{R}^3$ at that instant.
- The particle does not appear or disappear: it exists at all instants of Γ .

Key properties:

1. All points of Γ are real, independent of the observation instant t_0 .
2. Each instant t_0 defines an **observable slice**, $\Sigma(t_0)$.
3. Complete existence extends throughout the entire temporal history.

3. Discrete Foldings of Space-Time

Discrete transformations are defined:

$$T_n : \mathbb{R}^4 \rightarrow M_n, \quad n \in \mathbb{N}, n \geq 1$$

Verbalization:

- Each T_n takes the complete space-time and generates a new "fold" or symmetry, denoted M_n .
- Each T_n acts upon the entire worldline Γ , not just at one instant.
- Physical properties emerge from these folds.

3.1 Level n = 1

- Flat space-time, no foldings.
- The worldline exists, but no polarity or color.

3.2 Level n = 2: Temporal Folding

- A splitting over the temporal dimension is introduced: $t \mapsto t^+$ and $t \mapsto t^-$.

- This generates:
 - Matter/antimatter duality
 - Electric polarity $+/-$
 - $U(1)$ symmetry
- **Equation verbalization:**
 "Each time t splits into two branches, positive t and negative t , representing matter and antimatter."

3.3 Level n = 3: Spatial Folding

- Involves the three spatial directions x, y, z .
- Each direction generates a **discrete coupling mode** with the particle's temporal existence.
- These three modes correspond to the **three colors** of QCD: red, green, and blue.

Confinement: only the combination of all three modes reproduces the complete symmetry and is observable.

- **Topological restriction verbalization:**
 $\text{Sigma_complete}(t) = \text{Sigma_red}(t) \cup \text{Sigma_green}(t) \cup \text{Sigma_blue}(t)$
 "The union of the three color sections at each instant t is the physically observable form of the particle."

4. Correspondence with Symmetry Groups

Level n	Folded Over	Symmetry Group	Physical Manifestation
1	None	—	Flat space-time
2	Temporal dimension	$U(1)$	Electric charge, matter/antimatter
3	Three spatial dimensions	$SU(3)$	Color charge, confinement
4	Complete space-time	?	Possible unification or quantum gravity

- Each T_n is interpreted as a **principal bundle** with structural group G_n .
- The $SU(3)$ generators correspond to the three spatial dimensions x, y, z .

Verbalization:

- A principal bundle is an object that assigns to each point of $\mathbb{R}^4$ a set of fibers with symmetry G_n .
- The action of G_n on the fiber determines how physical properties (like color) manifest.

5. Formalization of Confinement

Let $\Sigma_i(t)$ be the spatial section of a quark with color c_i .

- Only the combination of all three colors is observable:

$$\Sigma_{\text{complete}}(t) = \Sigma_{\text{red}}(t) \cup \Sigma_{\text{green}}(t) \cup \Sigma_{\text{blue}}(t)$$

- **Verbalization:**

"Sigma complete at time t is the union of the red, green, and blue color sections. Only this combination can be physically observed; an isolated color cannot exist."

6. Consequences and Predictions

1. Geometric interpretation of color confinement.
2. Direct relationship between spatial three-dimensionality and color charge.
3. Coupling constants as topological invariants of the foldings.
4. Conceptual framework for extensions $n > 3$ and possible unification with quantum gravity.

7. Continuity and Discreteness

- The Gamma worldline is continuous in $\mathbb{R}^4$.
- The T_n foldings introduce **discrete levels** that generate symmetries and observable properties.
- The continuous (Gamma) and discrete (foldings) combine into an **arithmetic-geometric framework**.

8. Conclusion

- The particle is a **space-time cylinder**, real at all instants.
- Matter/antimatter, electric polarity, and color emerge from space-time geometry and its discrete foldings.
- The proposal unifies continuity and discreteness and offers a geometric interpretation of fundamental symmetries.

"Space does not contain matter; matter is geometry itself, folded along its entire temporal history."

Complete Narrative Framework: Particles, Antiparticles and Energy Reduction

1. Introduction

- A geometric particle framework is proposed where each particle is represented as a space-time cylinder in four real dimensions, not as a point.
- This framework describes temporal persistence and spatial interaction of particles and antiparticles continuously.
- The postulates are consistent with relativity and particle physics, offering a novel approach to modeling annihilation and energy transformation.

2. Fundamental Postulates

2.1 Worldline and Spatial Section

- Each particle has a worldline.
- The first particle has worldline called Gamma one. Gamma one consists of all quadruples of the form: position X, position Y, position Z, and time t, for all real values of t.
- The second particle has worldline Gamma two, defined analogously.
- The spatial section of the first particle at instant t is called Sigma one of t.
- The spatial section of the second particle at the same instant is called Sigma two of t.

- For almost all time t , Sigma one of t and Sigma two of t do not overlap, meaning no complete coincidence in three-dimensional space.

2.2 Energy Reduction of Spatial Structure

- If there exists a critical instant t_{critical} such that Sigma one of t_{critical} equals Sigma two of t_{critical} , **energy reduction of spatial structure** or **resolutive expansion of particular space** occurs.
- The spatial section transforms, releasing energy, while the temporal dimension persists.

2.3 Temporal Persistence Function f

- Function f takes the union of both particle worldlines and produces the remaining worldline, called $\Gamma_{\text{remaining}}$.
- This ensures temporal existence continues even when spatial structure transforms.
- Function f also determines how released energy is distributed along the remaining worldline.

2.4 Application to Particles and Antiparticles

- Particles and antiparticles are complementary structures.
- If Sigma of particle coincides with Sigma of antiparticle at t_{critical} , energy reduction of spatial structure occurs.
- This geometrically formalizes matter-antimatter annihilation and energy release.

3. Released Energy

- Let ρ be the energy density associated with the spatial section at critical instant t_{critical} .
- Local energy at critical instant is obtained by summing all energy density in the coincident spatial section.
- For particles with masses m_{one} and m_{two} , energy density approximates sum of masses multiplied by speed of light squared, concentrated at coincidence position.
- Total released energy approximates sum of masses times c^2 : released energy approximately equal to $(m_{\text{one}} + m_{\text{two}}) \times c^2$.

Narrative interpretation:

- All particle mass converts to energy at instant of spatial coincidence.

4. Temporal Energy Distribution

- Function f allows distributing released energy along remaining worldline.
- Before critical instant t_{critical} , released energy is zero.
- After t_{critical} , function f indicates what fraction of energy releases at each instant.
- This enables modeling progressive or gradual energy release over time.

5. Consequences and Contributions

1. Geometric formalization of particles and antiparticles as space-time cylinders, including complete temporal history.
2. Spatial exclusion with temporal persistence, avoiding complete overlap while maintaining temporal continuity.
3. Energy reduction of spatial structure, offering geometric framework for matter-antimatter annihilation.
4. Temporal persistence function f , novel for modeling energy redistribution over time.
5. Geometric energy release estimation, consistent with E equals m times c squared.
6. Conceptual application to high-energy physics and cosmology, useful for extreme collisions and high densities.

6. Potential Impact

- This framework provides a **new conceptual and geometric approach** to theoretical physics.
- Could inspire advanced simulations of particle-antiparticle interaction considering complete space-time cylinders.
- Offers tools for exploring extreme density limits, fusion, energy redistribution, and temporal continuity in particle systems.

7. Conclusion

- Relativity, energy conservation, and particle theory combine into a **rigorous and novel geometric model**.
- Clear postulates introduced: spatial exclusion, energy reduction of spatial structure, and temporal persistence.
- Directly connects with known physical phenomena, but provides **a more complete geometric interpretation**, accessible for future theoretical research.